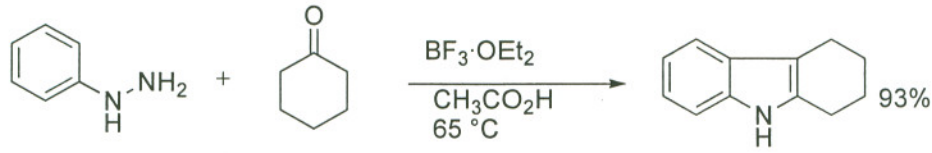
Organic Cumulative Exam, September 26, 2008

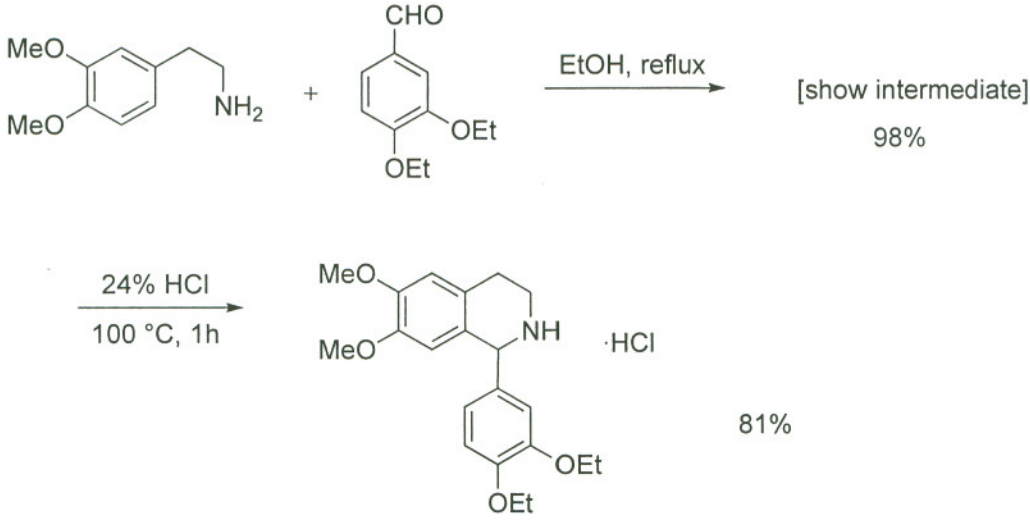
The following reactions are classical methods of heterocycle synthesis that employ simple carbonyl chemistry. All of these reactions were developed in the 19th and early 20th century. Note that all of them are, in modern terminology, 'green' and highly 'atom-economical'. The following examples were taken from a textbook published in 1968 (L. Paquette, *Modern Heterocyclic Chemistry*, W. A. Benjamin: New York).

Draw a reasonable mechanism for each one.

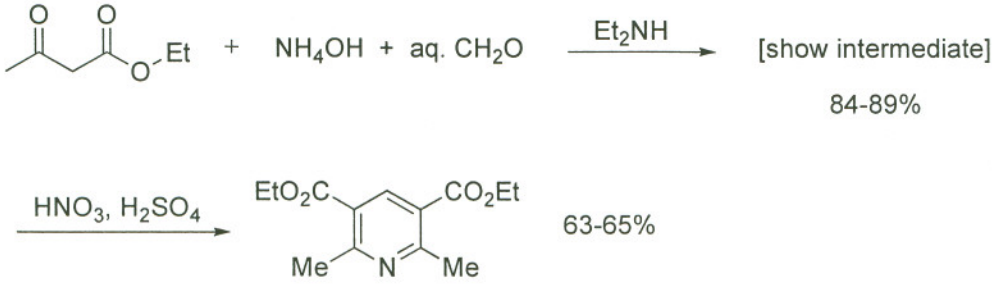
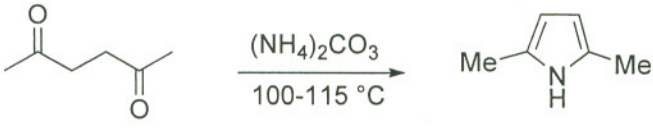
1. The Fischer Indole Synthesis



2. The Pictet-Spengler tetrahydroisoquinoline synthesis



3. The Hantzsch pyridine synthesis



4. The Paal-Knorr pyrrole synthesis